

Work with percentage change, growth and decay

Solve problems involving percentage increase, decrease and compound change.

Step 54

★ Today I am learning to work with percentage change, growth and decay.

1. Percentage Increase

Find the new value after the percentage increase.

- a) Increase 60 by 20%.
Answer: _____
- b) Increase 150 by 35%.
Answer: _____
- c) Increase 80 by 12.5%.
Answer: _____
- d) Increase 240 by 15%.
Answer: _____

- e) Increase £75 by 25%.
Answer: _____
- f) Increase 1.2 kg by 50%.
Answer: _____
- g) Increase 90 m by 8%.
Answer: _____
- h) Increase 360 by 5%.
Answer: _____

Think about it!

- To increase by $p\%$, multiply by $(1 + \frac{p}{100})$.
- 20% increase means multiply by 1.20.
- $12.5\% = \frac{12.5}{100} = \frac{1}{8}$.

2. Percentage Decrease

Find the new value after the percentage decrease.

- a) Decrease 80 by 20%.
Answer: _____
- b) Decrease 200 by 15%.
Answer: _____
- c) Decrease 90 by 5%.
Answer: _____
- d) Decrease 500 by 12%.
Answer: _____

- e) Decrease £120 by 30%.
Answer: _____
- f) Decrease 2.5 kg by 40%.
Answer: _____
- g) Decrease 150 m by 25%.
Answer: _____
- h) Decrease 600 by 8%.
Answer: _____

Think about it!

- To decrease by $p\%$, multiply by $(1 - \frac{p}{100})$.
- 15% decrease means multiply by 0.85.
- You can also find $p\%$ of the original and subtract it.

3. Percentage Change (Increase or Decrease)

Find the new value after the percentage change.

- a) Increase 70 by 10%.
Answer: _____
- b) Decrease 90 by 10%.
Answer: _____
- c) Increase 160 by 25%.
Answer: _____
- d) Decrease 240 by 25%.
Answer: _____

- e) Increase £200 by 12%.
Answer: _____
- f) Decrease £75 by 12%.
Answer: _____
- g) Increase 45 by 5%.
Answer: _____
- h) Decrease 45 by 5%.
Answer: _____

Remember:

- A 10% increase followed by a 10% decrease does not return to the original value.
- The order of increases and decreases can give different results.

4. Compound Growth

Find the value after the given annual growth.

- a) An investment of £1,000 grows by 5% each year. What will it be worth after 3 years?
Answer: _____
- b) A population of 2,000 increases by 3% each year. What will it be after 4 years?
Answer: _____

- c) The value of a car is £12,000 and increases by 4% each year. What will it be worth after 3 years?
Answer: _____
- d) A savings account grows by 6% each year. How much will £800 be worth after 2 years?
Answer: _____

Think about it!

- Each year, the growth is applied to the new value.
- Multiply by $(1 + \frac{r}{100})$ each time.
- r = growth rate per year.

5. Compound Decay

Find the value after the given annual decay.

- a) A value of £5,000 decreases by 10% each year. What will it be worth after 3 years?
Answer: _____
- b) A radioactive substance has a mass of 100 g and decays by 15% each year. What will be left after 2 years?
Answer: _____

- c) The value of a machine is £8,000 and decreases by 8% each year. What will it be worth after 4 years?
Answer: _____
- d) A population of 5,000 decreases by 2% each year. What will it be after 5 years?
Answer: _____

Think about it!

- Each year, the decay is applied to the new value.
- Multiply by $(1 - \frac{r}{100})$ each time.
- r = decay rate per year.

6. Mixed Problems

Solve each problem.

- a) The price of a jacket is £80. It is increased by 15% in January and then decreased by 10% in February. What is the price after February?
Answer: _____

- b) A phone that cost £600 decreases by 20% in value each year. How much is it worth after 2 years?
Answer: _____

- c) An investment grows by 7% in the first year and by 5% in the second year. If the initial amount is £2,000, what is the value after 2 years?
Answer: _____

- d) A laptop loses 25% of its value in the first year and 15% of its value in the second year. If it originally cost £1,200, what is it worth after 2 years?
Answer: _____

Remember:

- For mixed changes, apply them one after the other.
- Write a factor for each change and multiply.
- Round only at the end if needed.

7. Real-World Problems

Solve each problem.

- a) A shop increases the price of a toy by 15%. A month later, it decreases the new price by 10%. The original price was £25. What is the price after both changes?
Answer: _____

- b) The number of subscribers to a channel increases by 12% each year. It currently has 15,000 subscribers. How many will it have after 3 years?
Answer: _____

- c) A chemical loses 5% of its mass each year due to evaporation. If it starts with 500 g, how much will remain after 4 years?
Answer: _____

- d) The value of a painting increases by 8% each year for 2 years and then decreases by 6% the following year. It was originally valued at £3,000. What is its value now?
Answer: _____

Think about it!

- Identify whether the situation is increase, decrease, growth or decay.
- Decide if it is one change or multiple changes.
- Use appropriate factors.

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Understands the concept ☐ Explains thinking clearly
- ☐ Applies methods accurately ☐ Needs more practice
- ☐ Solves problems confidently

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____